1. What is $\frac{d}{dx}\left(\sqrt{2x+1}\right)$?

A. $f'(4) =$ _____

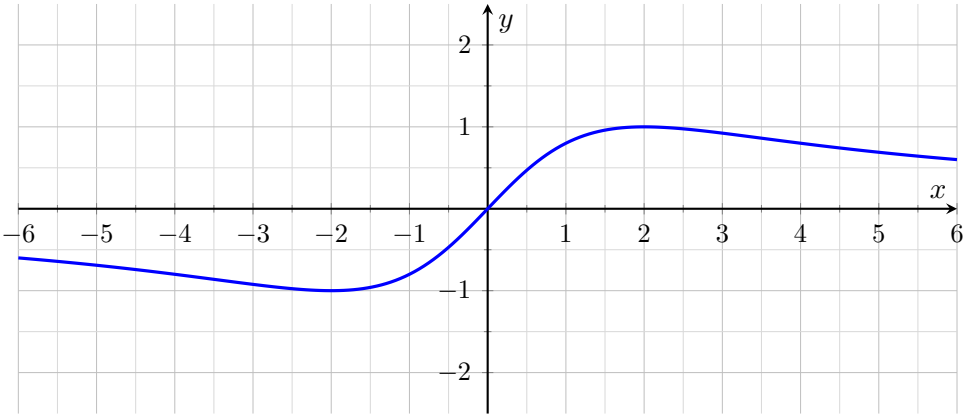
2. Sketching the graph of a derivative.

A. Study the graph of $f(x)$ shown below. Pay attention to where the function is increasing, decreasing, and where it has horizontal tangent lines.

B. In the empty grid below, sketch the derivative $f'(x)$.

C. Remember:

- Where $f(x)$ has horizontal tangent, $f'(x) = 0$
- Where $f(x)$ is increasing, $f'(x) > 0$ (above the x-axis)
- Where $f(x)$ is decreasing, $f'(x) < 0$ (above the x-axis)
- Where are there inflection points



Sketch the derivative f′(x) below:

